

Hoang Minh Thanh

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Passionate about Digital Humans, WebGL, and real-time rendering. I combine a strong foundation in machine learning with hands-on expertise in 3D human modeling. My contributions to DeepGesture and the GENE Challenge demonstrate full-stack capabilities, from training transformer and diffusion models to deploying evaluation pipelines and interactive 3D systems using Python, C++, and WebGL. I aim to bring both research rigor and engineering execution to advance research development.

Education

AS	University of Science - VNUHCM, Information Technology	2012 – 2015
BS	University of Science - VNUHCM, Computer Science	Sept. 2018 – Sept. 2020
	- GPA: 3.3/4.0 Thesis: GCAT - Link Prediction in Knowledge Graph [Report] [Code] - Advisor: PhD. Le Ngoc Thanh	
MSc	University of Science - VNUHCM, Computer Science	Oct. 2021 – Dec. 2024
	- GPA: 3.44/4.0 Thesis: OpenHuman: A conversational gesture synthesis system based on emotions and semantics [Report] [Code] - Advisor: Assoc. Prof. Dr. Ly Quoc Ngoc	

Publications

DeepGesture: A conversational gesture synthesis system based on emotions and semantics [arXiv] [homepage]	Jul. 2025
Towards a GENE Leaderboard [arXiv] [homepage]	Oct. 2024
Graph Collaborative Attention Network for Link Prediction in Knowledge Graphs [arXiv] [homepage]	Oct. 2020

Experience

Rakumo Inc, Software Engineer	Jul. 2019 – Apr. 2020
Developed a Django-based web application with integrated DocuSign.	
FPT Software, AWS Data Engineer	Oct. 2020 – Nov. 2021
- CI/CD data pipeline terraform in Jenkins, implemented and modified real-time and batch data pipelines on AWS using Kinesis, Lambda, S3, Glue, StepFunction, SNS, ..etc.. - Cox Automotive (onsite-offshore): Collaborated with international teams in Agile	
VNG Corp, Software Engineer	Oct. 2021 – May 2024
ZDN Team (Zalo Content Delivery Network): caching service handle millions req	
- Implement scheduled threadpool for monitoring stats execution. - Integrate outlier detection for migrating DDoS within the caching server. - Develop dashboard to real-time update server status.	

- Join GENE Research Team ([GENEA Leaderboard](#), [HEMVIP](#)): Designed and built a full-stack, production-ready leaderboard system to evaluate multiple DeepLearning gesture generation models, integrating user study workflows via Prolific.

Projects

DeepGesture ([Code](#)): Diffusion model for generative human motion with Transformer-based. Integrate with DeepPhase-Unity. deepgesture.github.io

Knowledge Graph Attention Network ([Code](#)): Performs link prediction on knowledge graphs by training TransE models with an attention mechanism applied to n-hop neighborhood graphs. graphattention.github.io

MillionScope ([Code](#)): ChatGPT-like with real-time streaming responses millionscope.com

Web-base realistic human render ([Code](#)): load mesh, apply glsl material: shine, coat, roughness, reflection, diffuse map. renderengine.pages.dev

Pet projects:

- [Pucca Runner](#): pathfinding game Dijkstra's algorithm
- [Blendshape animation](#), [MetaPet](#), [cortexpod.com](#); [pithagon.com](#)

Certifications

- [Critical Thinking Certification 2014](#), [Emotion Recognition Certification 2019](#), [Summer ML](#)
- Coursera: [Design Pattern](#), [TensorFlow Developer Professional](#) 2021
- AWS: [Solutions Architect - Associate](#), [Machine Learning - Specialty](#) 2021
- [Solution Architect Certifications](#) 2021

Technologies

Expertise: transformation matrix, digital human, quaternion, representation learning, animation synthesis, real-time scalable system

Proficient: Python, Jupyter notebook, Web (React, Next, Vite, mdx.js, tailwindcss), WebGL, GLSL Shader.

Tools: MotionBuilder, Maya (blendshape, rigging), Blender (bones, shapekey, hair particle), Zbrush (Projection, Layers), RDS ZWrap (Transfer topology), Marmoset Toolbag, Unity, Photoshop

Experience: BRDF, Retarget animation, transfer facial expression, skeleton builder, photogrammetry